Data & AI Case Study

Mapped to two Wuerth internship roles

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This is my independent case study. I mapped the requirements of two Wuerth Data & AI internship postings to work I have already built and measured in my own portfolio (now 23 repositories, including the decision-chain integration capstone -- one real dataset through the whole distributor chain with 13 machine-checked reconciliation identities plus four additive ones -- an MCP agentic-integration server with contract-enforced result provenance and idempotent result caching, two warehouse-logistics flagships, a supply-network optimizer, an energy forecasting + dispatch study, a visual quality-inspection study with SPC monitoring, a fraud-operations study with gated retrain promotion, a predictive-maintenance policy study, and a live installable quantum-explainer PWA). The euro and accuracy figures throughout are on synthetic data (except retail-analytics-real and decision-chain, measured and labelled on real public data) and exist to prove the methods work and are measurable -- not to forecast Wuerth outcomes.

Job #1 -- (Agentic) Automation with Low-code Platforms

Agentic AI workflows, low-code (n8n / Power Automate), connecting systems & APIs, document automation, ROI, rapid prototyping.

Job #2 -- Data & AI Analytics

BI / Power BI, KPI dashboards, forecasting & predictive analytics, Python & SQL, data modelling, turning data into decisions.

Who Wuerth is (public info)

Wuerth (public profile, approximate): a large family-owned German-HQ group in assembly/fastening and industrial MRO distribution -- ~400+ companies in 80+ countries, ~87,000 employees, ~EUR 20B+ revenue, a catalogue in the millions of articles, multi-channel (field sales, branches, e-commerce, e-procurement/EDI, ORSY inventory systems). A business that runs on assortment, pricing, availability, logistics, and high-volume transactional documents -- a natural fit for applied Data & AI.

Opportunity map (summary)

Wuerth business area (public-info) -> Data/AI approach -> repo + measured result (synthetic unless noted)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
Procurement / assortment / cross-sell	MILP vs greedy; Apriori/FP-growth; rule redundancy pruning	revops-optimizer, market-basket-analysis	MILP > greedy, 284 rules, top-lift 2.41, 41% of rules redundant -- the 149 kept carry all the information
Pricing & margin	Elasticity + leakage; endogeneity fix; per-move robustness gate; ablation	revops-optimizer, sales-kpi-analytics, ml-models-lab	EUR2.6M leakage lever; bias +1.52 -> +0.03; price-move gate 17/20 ACCEPT -- the biggest move (145% of the pricing uplift) is HELD, stated as a screen; ablation: endogeneity control knockout +1.446 RMSE -- the load-bearing piece
Inventory / replenishment	Newsvendor + forecast + ROP/EOQ policy + supplier reliability	distributor-intel-platform, ml-models-lab	MASE 0.38; MASE 0.987 / RMSSE 0.948 (beats naive + Holt-Winters); policy EUR127,421 working capital (cycle+safety exactly), 5.5x turns, 99.9% fill; measured lead times cost +14.4% safety stock -- wobble is 3/4 of the bill
Sales KPIs & analytics	KPIs + rolling-origin CV + pacing interval + rep decomposition	sales-kpi-analytics	MASE < 1; exec PDF; pacing EUR10.91M = 95.7% of plan, 80% interval -- closed-year back-check landed above the band, reported; rep decomposition (indirect standardization): 92% of the league table is territory, 8% execution
Logistics / routing	OR-Tools CP-SAT VRP + robustness + Fleet Size and Mix VRP	route-optimizer	4.6% / 31% savings; 5% headroom; in failing scenarios 96% -> 44%, expected day -4.6%; fleet mix: -17.5% vs the status quo with the service price shown (longest route +25.2%)
E-procurement automation	Agentic + n8n low-code + reliability benchmark + dry-run flow cost estimator	agentic-automation-lab, agent-flow-studio	~EUR625k/yr modelled; 1,350 seeded trials; content faults fail silently; stated (assumed rates); flow estimator: ~EUR145k/yr modelled; engine-verified branch scenarios, not a bill (declared rates)
Document processing	Agentic extract + 7-rule validation gate + cost model	doc-extract-agent	combined-gate precision 70.0% -> 87.5%; cost model: auto-post pays only above 98.4% precision -- the gate alone would lose money (modelled); the pre-fill carries the value
Customer retention	Decline + churn classifiers + ablation	revops-optimizer, ml-models-lab	ROC-AUC 0.90; ECE 0.107 -> 0.021; bootstrap skill CI +0.457 [+0.381, +0.528]; ablation: freq. slope knockout -0.247; PR-AUC 0.407, 14/14 folds
Warehouse / WMS (new)	WMS twin + factory/plant sim (Story Mode, 894-element plant, definable objects, multi-way line sim + fluids solver); slotting + DES + packing + pick-path routing + order batching	logistics-flow-studio (WarehouseTwin v3.19), logistics-digital-twin (engine)	-48.6% pick travel; ABC ~21% > random; ISO 22400 KPIs; 112/112 self-test + 45 harnesses; line sim 112.5 parts/hr (~88.1% eff.); fluids solver (modelled); engine slotting -44.2% (golden-zone 25% -> 100%); fill 2.0% -> 30.2%; routing: return +3.0% vs exact optimum, optimized layout ~46% shorter; batching: savings -71.3%, routing flips to largest-gap
Supply-network design (new)	Facility MILP + flows + safety stock + service frontier + growth plan	supply-network-opt	-21.8% cost vs greedy; stock -65.7% / -80.1%; frontier 2,353 -> 14,750 per service point (6.3x); growth: +8.8% headroom, 4th DC first pays at 1.30x (planning estimates)
Energy management / facilities (new)	Load forecast (rolling CV) + peak-shaving LP + causal dispatch backtest + degree-day decomposition	energy-demand-forecast	MASE 0.497; 4 knockouts not counted as wins (Holt-Winters loses to naive, reported); peak -20.9%; ~EUR11,100/yr at ASSUMED tariff; battery does not pay back on these assumptions; backtest captures 72.7% of the LP bound (robust variant loses, reported); decomposition recovers designed balance points exactly, weather 4.5% of energy 13.8% identified; PASS (modelled attribution)
Quality inspection (new)	Clean-only anomaly detection; pre-registered rule; SPC p-chart + WE rules; measured OCAP	quality-anomaly-vision	AP 0.779 vs SPC AP 0.772 (ROC-AUC inside 0.02 margin, PPA recommended); TPR 0.407 vs 0.393 @ 5% FPR; calibration correction 0 -> 9.70% measured; OCAP: label-free re-centering is a trap (green chart, real drift); screen-refit on verified-clean frames recovers 99% of the drift cost
Real data (completed)	Cleaning + RFM + returns + lifecycle + leakage-safe CV	retail-analytics-real	real data: seasonal-naive wins CV MASE 1.094; returns 3.63% of gross; 95.0% of value matched, median 10-day lag; Champions 25% -> 69.0%; lifecycle: resurrections outnumber repeats (439 vs 390/month)
Chain integration & reconciliation (new)	Provenance-tagged pipeline + identity ledger; MCP agentic layer w/ machine-readable result provenance + idempotent caching	decision-chain, chain-mcp	(p), (q); per-order spread to the cent over 4,151 orders (top decile 59.0%, Gini 0.665); fleet knob: only the transport line moves, vans unpriced (model property, not fleet advice); cross-repo revenue GBP 19,643,861.62 to the penny; naive wins lumpy (MASE 1.782); contract-enforced provenance + cache-status, 193 tests
Fraud & transaction-risk ops (new)	Cost-based alerting; gated retrain promotion; selective-labels feedback sim	fraud-detection-ops	PR-AUC 0.270 vs oracle 0.367; swap-set gates -> PROMOTE at 8,632 vs 8,841 -- retrain finds no new fraud, wins by shedding load (stated); feedback sim: ranking survives censoring, probabilities + alerts collapse
Maintenance & asset reliability (new)	Censored Weibull + age-replacement vs condition-based + CBM re-tuning	predictive-maintenance	beta 4.81; T* = 44.4 d at 7.16/machine-day -- the calendar rule beats the repo's own detector at the default threshold, reported; re-tuned economically the ranking flips to CBM (4.02) -- bounded, in-sample

Job #1 -- (Agentic) Automation with Low-code

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

Requirement	Repo / artifact	Proof (synthetic unless noted)
Build AI-agent workflows -- and measure how they fail	agentic-automation-lab (+ eval/reliability.py benchmark)	agentic tool-use loops; reliability benchmark (1,350 seeded trials): content faults fail silently past the guards and retries can't zero them (assumed fault rates, stated); 51 tests
Agentic integration via an open standard (MCP)	chain-mcp (official mcp Python SDK; Claude Desktop / Code configs)	6 real engines as AI-callable tools; typed schemas; never-crash errors; contract-enforced machine-readable result provenance (data label + engine commit + determinism flag); idempotency + result caching -- the key binds args + the exact engine checkout, and every success says computed-or-replayed (provenance.cache, contract-required); 193 tests incl. live JSON-RPC handshake
Low-code (n8n / Power Automate)	agentic-automation-lab: n8n/rfq_intake_agent.json	RFQ-intake agent; ~EUR625k/yr
Visual / flow-based building	agent-flow-studio (+ estimator.js dry-run estimator)	flow builder; ~EUR47k/yr; dry-run cost/latency estimator: declared rate card, branch scenarios enumerated and proven against real engine runs branch-for-branch -- 'emphatically not a bill' (78 node tests)
Connecting systems / APIs	agentic-automation-lab connector layer	agents call external APIs
Process automation (back-office)	doc-extract-agent	RFQ/invoice -> structured
Document intake / understanding	doc-extract-agent + 7-rule validation layer + cost model (eval/run_cost)	extract + validate; ~EUR145k/yr; combined-gate precision 70.0% -> 87.5%; cost model: auto-post pays only above 98.4% precision -- the gate alone would lose money (modeled), the pre-fill carries the value
Rapid prototyping	agent-flow-studio, agentic-automation-lab, logistics-flow-studio	runnable prototypes incl. a full installable offline WMS twin + factory/plant simulator (PWA, v3.19; 45 modules + 112/112 self-test)
Prototyping + education (hype-free)	quantum-explainer -- LIVE: dimitres-kisimov.github.io/quantum-explainer; bio-efficient-ai (learned BioHash vs random, public MNIST benchmark)	hand-written simulator ~300 lines, zero deps; 42 physics assertions + 57 structural checks pass; BioHash: mid-range wins from a 10x smaller circuit -- negatives kept (both tips lost, no memory-frontier point, corpus-dependent)
Show automation ROI	automation-roi-explorer	EUR383k/yr net modelled
Python engineering	all automation repos	Python throughout
Stakeholder communication	automation-roi-explorer	business-readable ROI views

Job #2 -- Data & AI Analytics

Every posting requirement -> repo + artifact + measured proof (synthetic data unless labelled real data).

Requirement	Repo / artifact	Proof (synthetic unless noted)
One dataset through the whole chain (real data + labelled layers)	decision-chain (run artifact + offline dashboard; FAIL path per identity)	13/13 reconciliation identities PASS + additive (n), (o), (p), (q); cross-repo revenue GBP 19,643,861.62 to the penny; ledger to the cent; forecast-error elasticity == holding share exactly (0.0580); per-order cost-to-serve to the cent over 4,151 real orders -- top decile 59.0%, Gini 0.665 (model shape, not profitability); fleet knob swept with every day's CVRP re-solved: only transport moves, vans unpriced (not fleet advice); naive wins lumpy (MASE 1.782); slotting optimum -1.6% vs ABC; CVRP -0.2% vs Clarke-Wright; cost rates labelled INVENTED
BI / Power BI	revops-optimizer: powerbi/DAX_measures.md + kpi_uplift_risk	P10/P50/P90 risk table w/ DAX
KPI dashboards / metrics + fair performance measurement	sales-kpi-analytics + pacing bullet chart + repperf.py	KPI layer + exec PDF; pacing EUR10.91M = 95.7% of plan, 80% interval; back-check: realised EUR11.28M landed above the band -- honest miss; rep decomposition (indirect standardization): 92% of the league-table dispersion is territory, only 8% execution
Uncertainty / risk quantification	revops-optimizer simulate.py + robustness.py; ml-models-lab BENCHMARK_CI.md	DAX pack + 3-page report + Monte-Carlo P10/P90 EUR145,691-170,723/yr, ~43% chance of clearing the headline; tornado ranks drivers; price-move gate: 17/29 ACCEPT, the biggest move HELD (67% carry) -- a screen, not a guarantee; bootstrap skill CIs: churn +0.457 [+0.381, +0.528]
Inventory policy / working capital	distributor-intelligence-platform dip/inventory.py + dip/reliability.py; supply-network-opt frontier	ROP/EOQ at ABC-XYZ targets: EUR127,421 working capital (cycle+safety exactly), 5.5x turns, 99.9% fill; supplier reliability: measured lead times cost +14.4% safety stock -- wobble is 3/4 of the bill, frontier 6.3x convexity
Predictive analytics / forecasting	sales-kpi-analytics; ml-models-lab global forecaster	MASE 0.38 [illegible]
Forecasting rigour + reconciliation guard	distributor-intelligence-platform (MRO command center)	[illegible] verdict over 16/16 cross-engine identities with all 21 headline numbers present
Energy forecasting + optimization	energy-demand-forecast (forecasters + LP + dispatch backtest + degree-day decomposition; 72 tests)	MASE 0.497 / MAPE 4.8%, 14/14 folds (Holt-Winters loses to naive reported); peak 368.2 -> 291.1 kW (-20.9%); ~EUR11,100/yr at ASSUMED EUR12/kW-month; timer baseline EUR0; battery does not pay back on these assumptions; causal backtest captures 72.7% of the LP bound (robust variant loses); decomposition recovers designed balance points exactly -- weather 4.5% of energy, 18% of the July peak (modelled attribution, not a sub-meter)
Python for analytics	sales-kpi-analytics, revops-optimizer	end-to-end pipelines
SQL / data modelling	sales-kpi-analytics SQL set	spend analysis queries
Excel-level tabular analysis	sales-kpi-analytics	spend breakdowns
Optimization / prescriptive	revops-optimizer; supply-network-opt; logistics-digital-twin	~EUR160k/yr; -21.2% cost vs greedy; fill 2.0% -> 30.2% (CP-SAT proof)
Elasticity / statistical modelling	revops-optimizer; ml-models-lab (+ ablation -> docs/ABLATION.md, drift-gated)	ROC-AUC 0.99; elasticity bias +1.52 -> +0.03; churn ECE 0.197 -> 0.021, bootstrap skill CI +0.457 [+0.381, +0.528]; ablation: freq_slope knockout -0.247 PR-AUC, endogeneity-control knockout +1.446 RMSE; the calibration knockout's bit-identical PR-AUC is the kept honest exception, null knockouts not counted as wins (numpy only, CI-reproducible)
Data into business decisions	sales-kpi-analytics; market-basket-analysis	EUR2.6M repackage lever; cross-sell [illegible]
Market-basket / cross-sell	market-basket-analysis + affinity.py + redundancy.py	[illegible]
Classification / anomaly detection	ml-models-lab; fraud-detection-ops	[illegible]
Model lifecycle / MLOps (gated retrain)	fraud-detection-ops challenger.py + feedback.py	[illegible]
Reliability / maintenance (survival)	predictive-maintenance pdm/policy.py + pdm/cbm_tuning.py	[illegible]
Visual quality inspection (vision)	quality-anomaly-vision (3 detectors, pre-registered rule; SPC + OCAP; 59 tests)	[illegible]
Warehouse / intralogistics + factory / process industry	logistics-flow-studio (WarehouseTwin v3.19); logistics-digital-twin (engine)	[illegible] optimized layout ~46% shorter; batching: savings -71.3%, routing flips to largest-gap (+1.2%)
Logistics / ops analytics	route-optimizer (46 tests, fleetmix.py); supply-network-opt (71 tests, growth.py)	[illegible] the service price shown (longest route +25.2%); a genuine mix wins the small instance (-8.5%; illustrative catalogue); safety stock -65.7% / -80.1%; frontier 6.3x; growth plan: +8.8% headroom, 4th DC first pays at 1.30x (planning estimates)
Real, messy data (completed)	retail-analytics-real (UCI Online Retail II, 1,067,371 real rows, CC BY 4.0)	real data: 94.0% rows kept; GBP 19.6M revenue; returns 3.65% of gross; 95.0% of value matched, median 10-day lag; Champions 25% -> 69.0%; lifecycle: resurrections outnumber repeats (439 vs 390/month); seasonal-naive wins CV, MASE 1.094
Present to decision-makers	revops-optimizer exec deck	exec narrative deck